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**CLASSIFICATION OF HYPERSPECTRAL IMAGES USING
MACHINE LEARNING AND DEEP LEARNING METHODS**

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SENIOR PROJECT

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We are very grateful to our advisor Dr. Ahmet ELBİR, who guided us throughout the project with his advice, patience and technical insight. But he was willing to go into the theory and the practice of what we were doing. He really made a difference.

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TABLE OF CONTENTS

LIST OF ABBREVIATIONS	vi
LIST OF FIGURES	vii
LIST OF TABLES	viii
ABSTRACT	ix
ÖZET	x
1 INTRODUCTION	1
1.1 Purpose	1
1.2 Preliminary Investigation	2
2 LITERATURE ANALYSIS	3
2.1 Traditional Machine Learning Approaches	3
2.2 Convolutional Neural Network-Based Approaches	3
2.3 Autoencoders, Generative Networks, and RNNs	4
2.4 Transformer Architectures and Semi-Supervised Methods	4
2.5 Current Standing and Original Contribution of This Study	5
3 SYSTEM ANALYSIS AND FEASIBILITY	6
3.1 System Analysis	6
3.2 Feasibility	7
3.2.1 Technical Feasibility	7
3.2.2 Legal Feasibility	7
3.2.3 Economic Feasibility	7
3.2.4 Workforce and Time Feasibility	7
4 SYSTEM DESIGN	9
4.1 Material and Dataset	9
4.1.1 Dataset Specifications	10
4.1.2 Data Structure and Storage	14
4.2 Methodology	14

4.2.1	Preprocessing and Dimensionality Reduction	14
4.2.2	Patch Extraction and Dataset Splitting	15
4.2.3	Model Architectures	15
4.2.4	Supervised Learning Models	16
4.2.5	Semi-Supervised Learning Models	16
4.2.6	Federated Learning Models	18
4.2.7	Comparative Learning Paradigm Summary	19
4.2.8	Evaluation Metrics	20
4.2.9	GUI and Multithreading Integration	21
5	EXPERIMENTAL RESULTS	22
5.1	Performance Analysis	22
5.1.1	Supervised Learning Results	22
5.1.2	Semi-Supervised Learning Results	23
5.1.3	Federated Learning Results	24
5.2	Comparative Analysis	25
5.2.1	Overall Accuracy Comparison	25
5.2.2	F1-Score Comparison	26
5.2.3	Performance Change Analysis	26
5.2.4	Architecture Comparison	27
5.2.5	Dataset-Specific Analysis	28
5.2.6	Privacy and Efficiency Trade-offs	28
6	CONCLUSIONS AND DISCUSSION	29
6.1	Summary	29
6.2	Key Findings	29
6.2.1	Semi-Supervised Learning Effectiveness	29
6.2.2	Federated Learning Privacy-Performance Trade-off	30
6.2.3	Architecture-Specific Insights	30
6.3	Project Objectives and Achievement Level	31
6.3.1	Original Objectives	31
6.3.2	Achievement Level	31
6.4	Strengths	32
6.4.1	Methodological Strengths	32
6.4.2	Implementation Strengths	32
6.5	Limitations and Weaknesses	33
6.5.1	Methodological Limitations	33
6.5.2	Experimental Limitations	33
6.5.3	Implementation Limitations	34
6.6	Future Work	34

6.6.1	Methodological Extensions	34
6.6.2	Architectural Enhancements	34
6.6.3	Experimental Directions	35
6.6.4	Practical Applications	35
6.7	Conclusion	36
	References	37
	Curriculum Vitae	39

LIST OF ABBREVIATIONS

1D-CNN	1-Dimensional Convolutional Neural Network
3D-CNN	3-Dimensional Convolutional Neural Network
AE	Autoencoder
R&D	Research and Development
CNN	Convolutional Neural Networks
GAN	Generative Adversarial Networks
GPU	Graphics Processing Unit
GUI	Graphical User Interface
HSI	Hyperspectral Images
MRF	Markov Random Fields
PCA	Principal Component Analysis
RF	Random Forest
RNN	Recurrent Neural Networks
SAE	Stacked Autoencoders
SVM	Support Vector Machines
ViT	Vision Transformer

LIST OF FIGURES

Figure 3.1 Project Gantt Chart (Feb 2026 - June 2026) 8

Figure 4.1 Overall Workflow of the Proposed HSI Classification System . . 9

Figure 4.2 Indian Pines – False-color band composite 10

Figure 4.3 Indian Pines – Ground truth map 10

Figure 4.4 Pavia University – False-color band composite 12

Figure 4.5 Pavia University – Ground truth map 12

Figure 4.6 Salinas Scene – False-color band composite 13

Figure 4.7 Salinas Scene – Ground truth map 13

Figure 4.8 General Block Diagram of the Proposed HSI Classification System 14

Figure 4.9 Flowchart of the Semi-Supervised Learning Strategy 20

LIST OF TABLES

Table 4.1	Groundtruth classes for the Indian Pines scene and their respective samples number	11
Table 4.2	Groundtruth classes for the Pavia University scene and their respective samples number	12
Table 4.3	Groundtruth classes for the Salinas scene and their respective samples number	13
Table 4.4	Patch counts after stratified 90/10 train-test split ($15 \times 15 \times 30$)	15
Table 4.5	Comparison of Supervised, Semi-Supervised, and Federated Learning Paradigms	20
Table 5.1	Supervised Learning Performance on Indian Pines Dataset	22
Table 5.2	Supervised Learning Performance on Pavia University Dataset .	23
Table 5.3	Supervised Learning Performance on Salinas Dataset	23
Table 5.4	Semi-Supervised Learning Performance on Indian Pines Dataset	23
Table 5.5	Semi-Supervised Learning Performance on Pavia University Dataset	24
Table 5.6	Semi-Supervised Learning Performance on Salinas Dataset . . .	24
Table 5.7	Federated Learning Performance on Indian Pines Dataset (Round 50)	24
Table 5.8	Federated Learning Performance on Pavia University Dataset (Round 50)	25
Table 5.9	Federated Learning Performance on Salinas Dataset (Round 50)	25
Table 5.10	Overall Accuracy Comparison Across All Learning Paradigms . .	25
Table 5.11	F1-Score Comparison Across All Learning Paradigms	26
Table 5.12	Average Performance Across Datasets by Architecture Type . . .	27
Table 5.13	Average F1-Score Across Datasets by Architecture Type	27
Table 5.14	Qualitative Comparison of Learning Paradigms	28
Table 6.1	Project Objectives and Achievement Level	32

ABSTRACT

CLASSIFICATION OF HYPERSPECTRAL IMAGES USING MACHINE LEARNING AND DEEP LEARNING METHODS

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This project formulates an integrated classification framework for hyperspectral images (HSI) covering supervised, semi-supervised, and federated learning methods. It particularly focuses on how each method tackles the challenge posed by high spectral dimensionality and low labeled data in the field of remote sensing. Central to the framework are three deep-learning models: 2D-CNN, 3D-CNN, and ViT. To reduce spectral data (which is typically larger than 200 spectral bands) down to 30 principal components, Gaussian filtering is first applied to the data to reduce the noise, followed by channel-wise normalization to reduce the noise between channels, and finally, through, PCA. The semi-supervised learning framework is achieved using a self-training framework within a curriculum learning framework where the confidence of self-predictions is used for ordering the unlabeled data, increasing the predictions' ordering confidence, and stabilizing optimization. A simulated server-client federated architecture enables privacy-respecting collaborative training and operates on the three benchmark datasets: Indian Pines, Pavia University, and Salinas. All models are evaluated using Overall Accuracy, Precision, Recall, F1 Score, and Cohen's Kappa coefficient. The results show a performance divide where, for the two larger datasets, semi-supervised learning achieved an accuracy of 86-97

Keywords: Hyperspectral Image Classification, Deep Learning, CNN, Vision Transformer, Semi-Supervised Learning, Federated Learning, Machine Learning, Curriculum Learning, Remote Sensing, Principal Component Analysis

HİPERSPEKTRAL GÖRÜNTÜLERİN MAKİNE ÖĞRENMESİ VE DERİN ÖĞRENME YÖNTEMLERİYLE SINIFLANDIRILMASI

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Bitirme Projesi

Danışman: Dr. Ahmet ELBİR

Bu çalışma, hyperspektral görüntü sınıflandırması için yapılandırılmış, yarı yapılandırılmış ve federasyonel öğrenmeyi entegre eden birleşik bir sistem geliştirmektedir. Öncelikle, uzaktan algılama alanında sıklıkla karşılaşılan iki probleme odaklanmaktadır: yüksek spektral boyut ve etiketli örnek eksikliği. Sistem, 2D-CNN, 3D-CNN ve Vizyon Dönüştürücü (ViT) olmak üzere üç derin öğrenme yapısına dayanır. 200'den fazla spektral görüntü bandı içeren karışık olmayan veriler, kanal bazlı normalizasyon ve Gaussian filtreleme sonrası İlk Bileşen Analizi (PCA) kullanılarak ilk 30 banda indirgenir. Yarı denetimli öğrenme bağlamında, kendi kendine eğitim mekanizması ve müfredat öğrenme yöntemleri kullanılır. Bu, sınıflandırılacak etiketlenmemiş örneklerin sıralamasının en basitten (yüksek güven) en zorlusuna (düşük güven) doğru ilerlemesini sağlar ve daha kararlı bir optimizasyon sağlar. Simüle edilmiş sunucu-client tasarımı, üç yaygın olarak tanınmış benchmark veri seti (Indian Pines, Pavia University ve Salinas) üzerinde federatif öğrenmeyi gerçekleştirirken gizlilik koruyan dağıtık eğitim sağlar. Modeller, geniş ve çeşitli performans ölçütleriyle değerlendirilir. Sonuçlar açık bir biçimde ayrımı ortaya koyar. Yarı denetimli öğrenme, büyük veri setlerinde rekabetçi performans elde eder;

Anahtar Kelimeler: Hiperspektral Görüntü Sınıflandırması, Derin Öğrenme, CNN, Vision Transformer, Yarı-Denetimli Öğrenme, Federe Öğrenme, Makine Öğrenmesi, Müfredat Öğrenmesi, Uzaktan Algılama, Temel Bileşen Analizi

1

INTRODUCTION

With the ability to capture electromagnetic signals across many narrow spectral bands, Hyperspectral imaging (HSI) reflects electromagnetic signatures across hundreds of spectral bands, revealing properties of the physical and chemical structures of the surface of the earth. Precise details at this level are not possible with conventional RGB imagery. HSI's capability to capture detailed spectral information make it highly useful in many fields including agriculture, environmental monitoring and defense. However, the advanced spectral information comes with the fundamental computational burden of the curse of dimensionality. When considering the scarcity of expertly annotated pixel samples, the cost of training machine learning and deep learning models increases significantly along with the model's generalizability.

The gap in the performance of supervised versus semi-supervised deep learning (CNN, ViT, Self-Training) architectures due to high dimensionality and the scarcity of labeled data is the main focus of this research work. Two primary things set this work apart. One, a curriculum learning approach is used to the training of these models in a bid to stabilize optimization. Two, a multi-threaded GUI was created to the point where users can load their own data, get predictions, and do all this without needing to code or even touch the command line.

The remaining chapters of this report review and analyze our work in these areas. Chapter 2 focuses on analyzing the state of the art research. Design and methodology of the entire system including data processing and architecture are discussed in Chapter 3. Chapter 4 provides a set of results based on quantitative comparative experiments and Chapter 5 provides analysis of these results and proposes ideas and recommendations for future enhancements.

1.1 Purpose

The primary aim of this project is to describe, in detail, the performance disparity between supervised and semi-supervised learning as applied to the classification of

high-dimensional hyperspectral data. The aim of this project can be broken into the following sub-aims:

- To evaluate the performance of supervised and semi-supervised models on the standard hyperspectral datasets: Indian Pines, Pavia University, and Salinas.
- To evaluate the impact of Gaussian filtering and PCA on the classification of hyperspectral data.
- To introduce Curriculum Learning in training to each model to improve training stability and convergence.
- To carry out a balanced and fair assessment of each model.
- To develop a Python multithreaded GUI, allowing end users to use the models to classify data in real time and with no need to access the code.

1.2 Preliminary Investigation

There is an observable shift in pattern from using traditional machine learning models towards using models built on deep learning architectures in the literature on hyperspectral image classification. Current models face problems when dealing with big data since most are trained using a lot of labeled data. Constructing labeled data for hyperspectral images is an expensive and time-consuming endeavor due to the need for an expert in the field to assist with the data labeling.

This project aims to fill important gaps in the literature. While most models translate well to theory, they end up being just a concept, since most of them are scripts that require technical expertise to use. The application, based on the experimental code, uses a that gives users the ability to input their own datasets and receive results in near real time.

2 LITERATURE ANALYSIS

Hyperspectral image (HSI) classification has been a focus area of remote sensing for many years, especially for being able to identify surface materials, something that broadband sensors are unable to do. There are challenges in the field that require a balance between deep models and the limited amount of training data that has been labeled. This section puts HSI classification methods into groups based on the models used and the challenges that they solve in order to help understand the development of these methods.

2.1 Traditional Machine Learning Approaches

In the beginning, HSI classification used shallow and statistical methods of machine learning. Dominant frameworks in this time were Support Vector Machines (SVM) and Random Forest (RF). An example of early work in HSI image classification was by Du et al. in using sparse discriminative representations and multinomial logistic regression to achieve 97.71

Although the methods mentioned previously are robust to noise and interpret output, they have become limited in terms of classification accuracy. A challenge for machine learning is to capture the associations in high-dimensional spectral spaces. Classical techniques in feature extraction also struggle to meet the accuracy of representations that deep networks learn.

2.2 Convolutional Neural Network-Based Approaches

Learning spatial and spectral features simultaneously and without manual feature engineering has changed HSI classification with CNNs. 3D-CNNs with L2 regularization and dropout were applied by Chen et al. and achieved a groundbreaking 99.66

A challenge of 3D-CNNs is the parameter cost. Therefore, some strategies were introduced. AcousticSN was able to balance spectral-spatial expressiveness against computational cost by combining both 2D and 3D convolutions. In a different approach, Ahmad et al. introduced small 3D-CNN architectures with faster inference times that maintained the same accuracy. Boundary artifacts were addressed by Paoletti et al. by mirroring borders, which mitigated information loss for the edges of each patch. Combined, these approaches provided a strong justification for deep learning features with CNNs for HSI classification, but their requirements for labeled data and GPU memory grow with depth, which limits their use in real-world situations with a lack of data.

2.3 Autoencoders, Generative Networks, and RNNs

Data that is insufficiently labeled has prompted the use of architectures that give representations of a decent quality without much supervision. For instance, Stacked Autoencoders (SAE) offer a good method of performing unsupervised feature extraction by dimensionality reduction. Combining SAEs with CNNs, as done by Li et al., focuses on learning spatial weights that are adaptive to one locality, while Zhao et al. focused on SAE-based compression and 3D deep residual networks that combine both dimensionality reduction and classification, but assume a significant amount of labeled data.

Sequence modeling in HSI began with Recurrent Neural Networks. In their work, Shi et al. employed RNNs to incorporate spectral-spatial features of different scales. They considered different spectral bands to be in a temporal sequence. Another approach to the problem was Generative Adversarial Networks, which synthesized samples and improved small labeled datasets and, in turn, aimed to mitigate the generalization issue caused by the lack of labeled samples.

Each of these frameworks imposes additional issues. SAE and RNN frameworks suffer from the burden of sequential computation, and update the systems slowly and GANs offer unpredictable results. Small changes to hyperparameters can annihilate generators or result in mode collapse and create uninformative training data.

2.4 Transformer Architectures and Semi-Supervised Methods

In the realm of HSI classification, Transformers improve on capturing long range spectral-spatial relations. Their addition to CNN based local feature extraction has also produced positive results. Recent tokenization methods referring to the encoding

of spectral-spatial areas as sequences have allowed more unmediated use of the Vision Transformer. Thanks to recent architectures featuring dual attention, benchmark results on various datasets have surpassed 99

Transformers have proven especially powerful in this field, but their increased architecture cost is not overshadowed by previous methods. Their cost and hardware demands are greater for large datasets and decrease their usefulness in low-data scenarios.

In the realm of semi-supervised learning, Sellami et al. showed that 3D deep networks, combined with adaptive band selection, leverage unlabeled data effectively and reduce the gap in accuracy of the labeled and unlabeled data. This research serves as an excellent proof of concept. Most of the problems this research identifies, like decision boundary bias regarding pseudo-label noise and confidence threshold selection, are still unresolved in the domain.

2.5 Current Standing and Original Contribution of This Study

A common theme in the reviewed research is that the models that are the most accurate when a lot of labeled data are available are also the most fragile models when labeled data are sparse, and vice versa. Transformers and deep CNNs tend to overfit when trained on small labeled datasets. Semi-supervised and generative approaches to the data problem introduced instability to the training process and sensitivity to thresholds, and no previous research has incorporated stability directly into the semi-supervised training process.

This research aims to solve this issue by combining CNNs and Transformers in a Self-Training framework incorporating Curriculum Learning. This is the technique that organizes the training samples in order of the estimated level of difficulty so that the model does not focus on the noisy pseudo-labels early on in the training process. In addition to its novelty, most of the other research in the domain consists of models that only function through command-line interfaces, which makes them inaccessible to most practitioners that do not have a specialized knowledge of deep learning. The work presented here aims to convert research models into applications that can be deployed and used outside of an academic environment through the introduction of a multithreaded PyQt6 GUI. It also aims to provide real-time predictions using other external datasets.

System analysis helps bridge design and problem identification. It tells us what must be done, the limitations and constraints the system will be bound within, and the requirements for measuring the quality of the outputs the system produces. In this chapter, project goals are described, the methods and processes of requirements elicitation are detailed, and the proposed solution is examined for feasibility in terms of technology, law, economy, and labor.

3.1 System Analysis

As this is an R&D project, requirements were not provided because there were no client specifications. Instead, requirements were developed through the analysis of issues and constraints provided by existing deep learning based HSI classifier algorithms. The main function of the system is to read and handle high-dimensional spectral data (either introduced externally by the user or extracted from established benchmarks - Indian Pines, Pavia University, and Salinas) and classify high accuracy pixel data.

The system is governed by three functional requirements:

- Raw hyperspectral cubes must be filtered using a Gaussian filter and PCA [1].
- Supervised (CNN [2], ViT [3]) and semi-supervised (Self-Training [4]) models must be trained in a Curriculum Learning framework [5] that regulates the difficulty of samples.
- A multithreaded GUI must be used to encapsulate all trained models, allowing users to perform simultaneous model testing and view per pixel predictions in real-time without modifying the underlying source code.

Meeting these requirements will be evaluated using Accuracy, Precision, Recall, F1-Score, and Cohen's Kappa coefficient.

3.2 Feasibility

The analysis examines the likelihood of the proposed solution in four categories: technical, legal, economic, and workforce.

3.2.1 Technical Feasibility

The proposed programming language is Python [6] since it has a complete ecosystem for machine learning with a favorable community. Deep learning is constructed with PyTorch [7] or TensorFlow [8] with data management offered by Scikit-learn [1] and OpenCV [9]. The design of the graphical user interface is done with PyQt6. The choice of the PyQt6 library was made because of its built-in support for multithreaded worker patterns, which is important for keeping the user interface responsive when time-consuming inference operations are performed. The construction is done with VS Code [10] and the versioning is done with GitHub. [11].

3.2.1.1 Hardware Feasibility

Training deep models with long inference operations on complete hyperspectral cubes requires high computational resources. Google Colab [12], provides the needed graphic processing units and eliminates the need for the computational resources for the project. The graphic user interface and integration design are done on the project team's personal macOS systems.

3.2.2 Legal Feasibility

All libraries in the stack of PyQt6, Python, PyTorch, and OpenCV are fully open-sourced. The three benchmark datasets in use are legally permitted for academic use. There are no licensing or patent barriers.

3.2.3 Economic Feasibility

The entire tool chain is free and open-source with Google Colab offering the graphic processing units at no cost. The project makes no additional capital investments with the teams' personal systems.

3.2.4 Workforce and Time Feasibility

The project team of Tan Erciyas and Kerem Baydar has the required skills to execute the entire project.

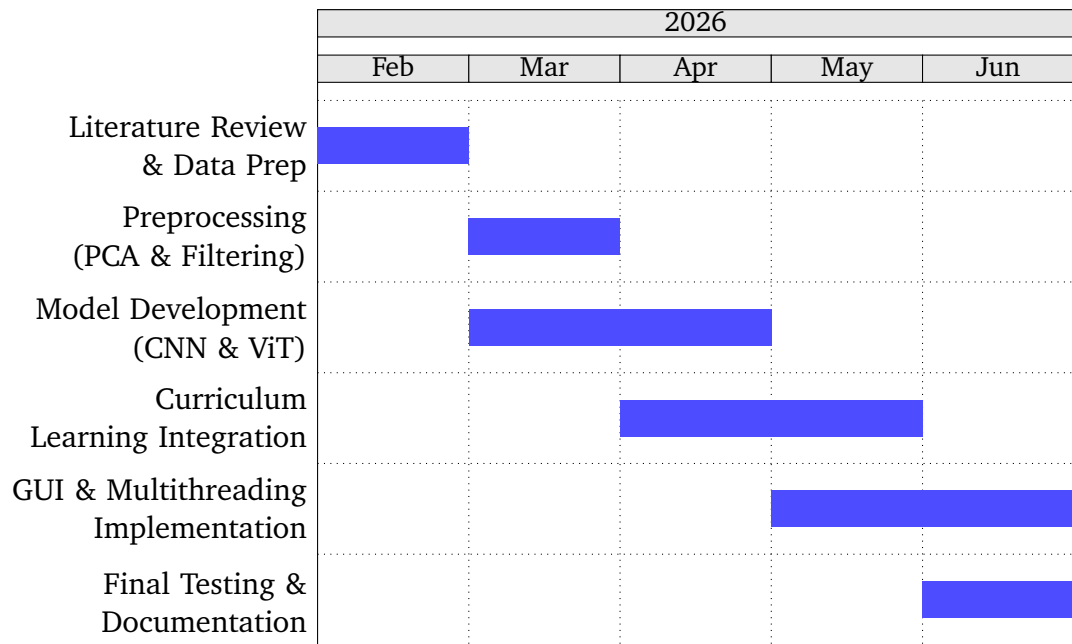


Figure 3.1 Project Gantt Chart (Feb 2026 - June 2026)

4 SYSTEM DESIGN

Translating requirements into an architectural framework occurs in system design. This chapter describes the datasets, preprocessing pipelines for spectral data, the three system core deep learning architectures, and the supervised, semi-supervised, and federated training methods examined.

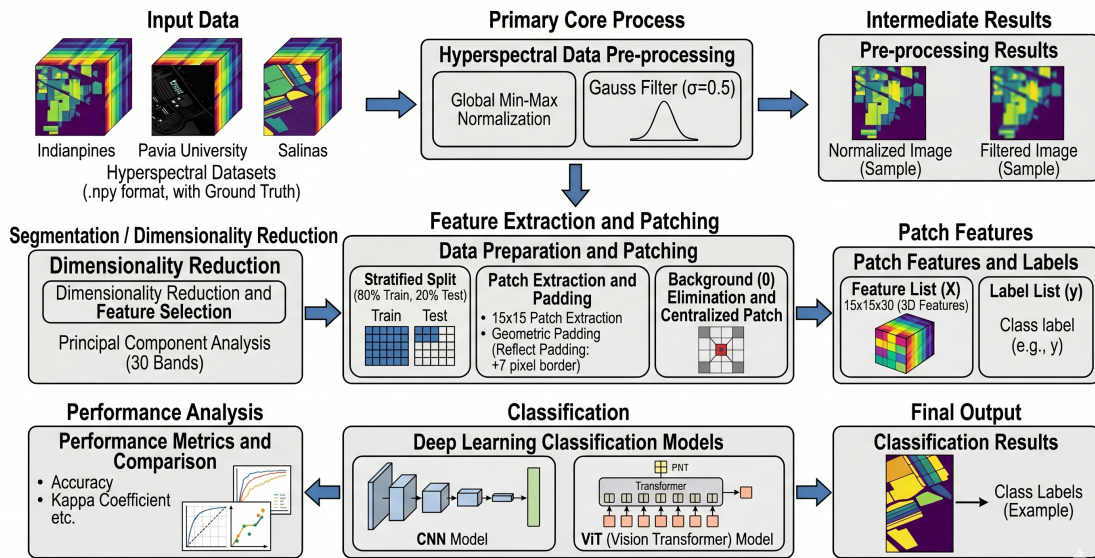


Figure 4.1 Overall Workflow of the Proposed HSI Classification System

4.1 Material and Dataset

The performance of different models on a classification task in hyperspectral imaging (HSI) is highly dependent on the different features of the used dataset. These features include the model's spatial resolution, the level of class imbalance, and the number of samples in the dataset with labels. In this work, we use three benchmark datasets from the remote sensing literature, and each of these datasets has its own set of challenges.

4.1.1 Dataset Specifications

4.1.1.1 Indian Pines

This dataset is collected by the AVIRIS sensor and has a 145x145 grid with 224 spectral bands on 16 vegetation and land-cover classes. In most cases, the water absorption bands reduce the number of usable bands to 200.

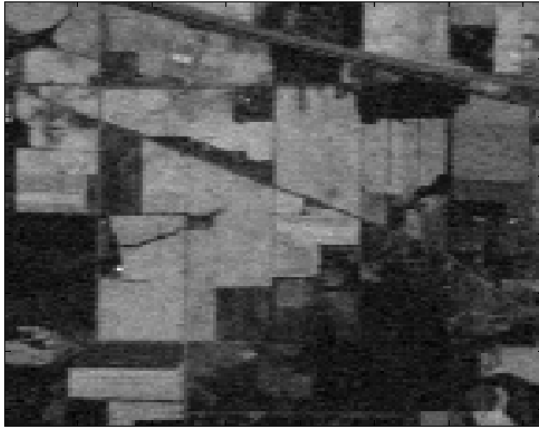


Figure 4.2 Indian Pines – False-color band composite

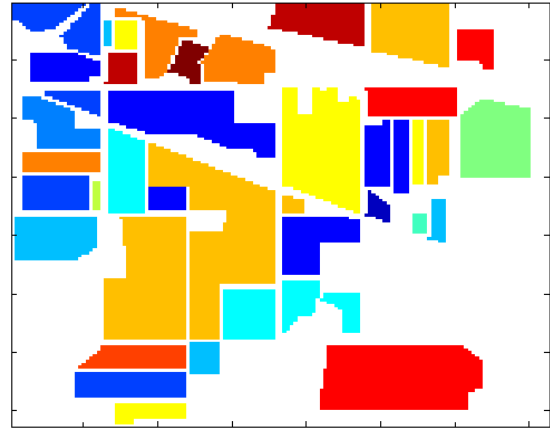


Figure 4.3 Indian Pines – Ground truth map

Table 4.1 Groundtruth classes for the Indian Pines scene and their respective samples number

#	Class	Samples
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
Total		10249

4.1.1.2 Pavia University

This dataset is collected by the ROSIS sensor and covers 610x340 pixels with 103 spectral bands. This dataset has 9 urban land-cover classes which makes this dataset a very challenging multi-class urban benchmark.



Figure 4.4 Pavia University – False-color band composite

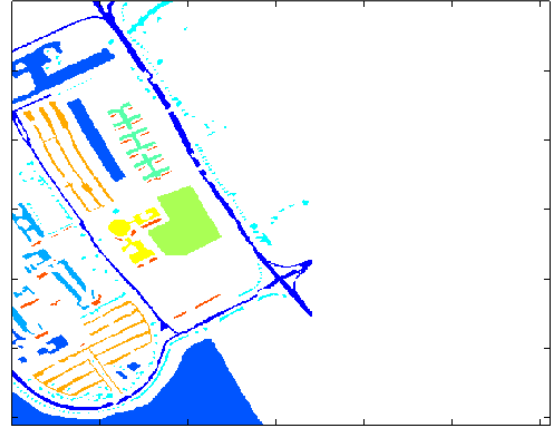


Figure 4.5 Pavia University – Ground truth map

Table 4.2 Groundtruth classes for the Pavia University scene and their respective samples number

#	Class	Samples
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Painted metal sheets	1345
6	Bare Soil	5029
7	Bitumen	1330
8	Self-Blocking Bricks	3682
9	Shadows	947
Total		42776

4.1.1.3 Salinas Scene

This dataset is also collected by the AVIRIS sensor but this time over the Salinas Valley of California. This dataset has a spatial resolution of 512x217 pixels which is also the highest resolution of the other two datasets. This dataset also contains 224 spectral bands which are reduced to 204 after removal of the water absorption bands.

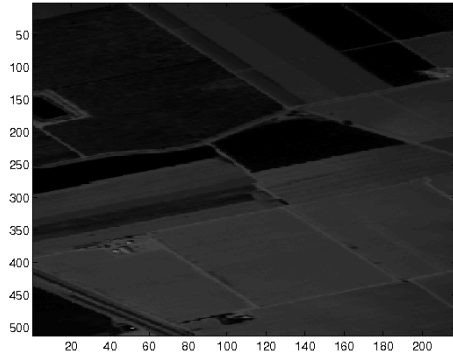


Figure 4.6 Salinas Scene – False-color band composite

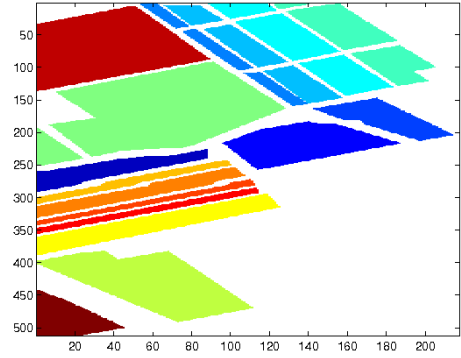


Figure 4.7 Salinas Scene – Ground truth map

Table 4.3 Groundtruth classes for the Salinas scene and their respective samples number

#	Class	Samples
1	Brocoli_green_weeds_1	2009
2	Brocoli_green_weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807
Total		54129

4.1.2 Data Structure and Storage

All of these datasets have the same 3D structure and can be visualized as a hypercube, where the dimensions of the hypercube are the spatial dimensions (H and W) and the number of spectral bands (B). All datasets are stored as NumPy arrays (.npz) and are loaded in spatial patches to facilitate the training process. This supports the local spatial context and allows us to use the limited memory efficiently.

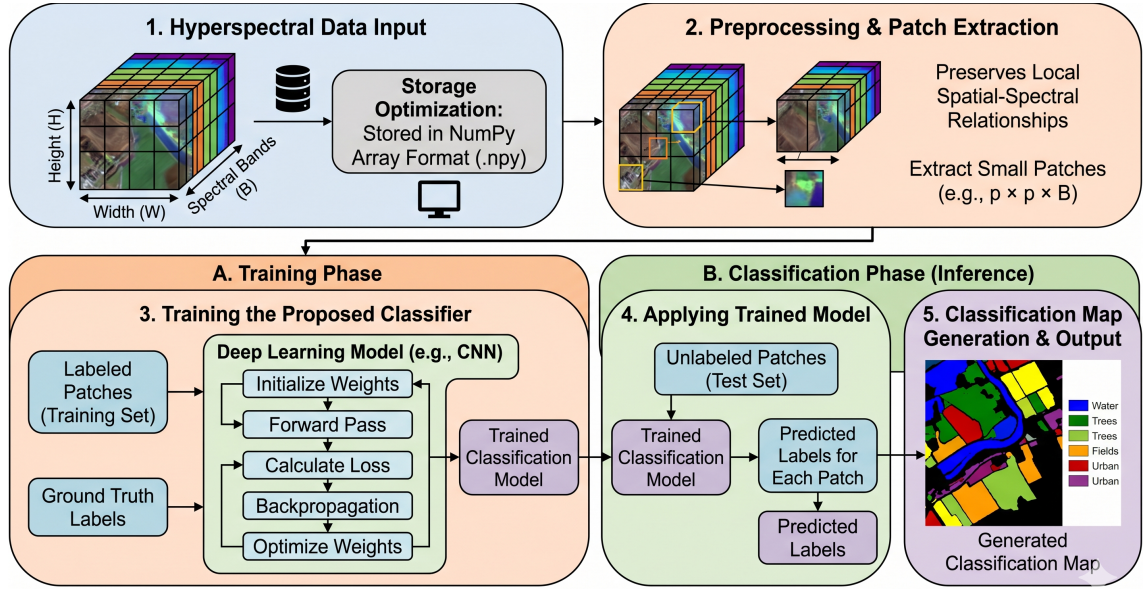


Figure 4.8 General Block Diagram of the Proposed HSI Classification System

4.2 Methodology

The processing pipeline consists of three sequential phases: preprocessing, patch-based feature extraction, and learning. Each phase is connected to the next, and the characteristics of hyperspectral data are reflected in the design of each phase.

4.2.1 Preprocessing and Dimensionality Reduction

Hyperspectral cubes are made of hundreds of noises and correlated bands and can mislead learned features. Consequently, the cost of training can increase greatly. In order to mitigate this, three operations are done.

- 1. Global Min-Max Normalization:** In this step, the hyperspectral data is reshaped from a 3D hypercube to a 2D matrix. From here, the global minimum and maximum across all pixels and bands of the reshaped data are calculated. Each value is then rescaled per the new global minimum and maximum.

Normalization of the bands was avoided to prevent variance loss.

$$\hat{x} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (4.1)$$

2. **Spatial Gaussian Filtering:** In this step the data is filtered using a Gaussian Filter. The value of σ was set to 0.5 and was only applied to the spatial axes of the 3D hypercube. A value of zero was set for the spectral σ ensuring that the relationships between the bands of the data were preserved.
3. **Fixed-Component PCA:** In this step, PCA is applied with a value for $n_components$ set to 30. Prior to this, a variance threshold of (99.5%) was applied. However, for both the Salinas and Pavia University datasets, this resulted in 4-5 PCA components. This was insufficient for the input depths of the Convolutional Neural Networks and Vision Transformers. The processed hyperspectral data was then reshaped to $(H, W, 30)$ [1].

4.2.2 Patch Extraction and Dataset Splitting

From each pixel (r, c) where $gt[r, c] > 0$, a 15×15 patch is sampled from the PCA cube, resulting in a sample of size $15 \times 15 \times 30$. All background pixels (label = 0) are omitted. A cube is padded $\lfloor 15/2 \rfloor = 7$ pixels on each side with reflective padding, allowing boundary pixels to be valid patch centers.

Patches are randomly split into training and test sets with `test_size=0.1`, `stratify=y`, and `random_state=42`. The resulting sample counts are given in Table 4.4.

Table 4.4 Patch counts after stratified 90/10 train-test split ($15 \times 15 \times 30$)

Dataset	Total	Train (90%)	Test (10%)
Indian Pines	10249	9224	1025
Pavia University	42776	38498	4278
Salinas Scene	54129	48716	5413

4.2.3 Model Architectures

Two model architectures represent different yet complementary approaches to learning spectral and spatial features:

- **3D-CNN:** With 3D convolution, spectral and spatial features are extracted

simultaneously. The model is designed with convolutional blocks with batch normalization and ReLU activations. [2].

- **Vision Transformer (ViT):** With a ViT, HSI patches are first linearized, then multi-head self-attention is applied to each spectral-spatial token. [3].

4.2.4 Supervised Learning Models

All models in this paradigm learn only from the labeled portion of the training dataset. It is also the baseline model for measuring the performance of semi-supervised and federated models.

4.2.4.1 2D-CNN Supervised Model

The 2D-CNN model uses $15 \times 15 \times 30$ patches and treats the 30 PCA components as channels. Convolutions in this model are applied over the spatial dimensions.

4.2.4.2 3D-CNN Supervised Model

The 3D-CNN model uses volumetric convolutions that simultaneously span both the spatial and spectral dimensions.

4.2.4.3 Vision Transformer (ViT) Supervised Model

In the ViT, HSI patches are first linearized and then augmented with learnable positional encodings to retain their order [3].

4.2.5 Semi-Supervised Learning Models

The collection of expert labeled pixel annotations is a long and costly process. Semi-supervised learning employs the extensive, yet unlabeled, pixel set as a training signal. The strategy combines self-training and curriculum learning to add to the labeled set progressively while containing the noise of pseudo-labels.

4.2.5.1 Self-Training Strategy

Pseudo-labeling self-training occurs in the following iterative manner:

1. A small, labeled training sample set is used to initialize the model.

2. Subsequently, predictions with confidence score are generated for the unlabeled sample set.
3. The sample set is then augmented by the inclusion of highly confident predictions.
4. The model is retrained on the augmented sample set.
5. The process is repeated until a convergence criterion is met [4].

4.2.5.2 Curriculum Learning Integration

Curriculum Learning (CL) helps structure the training procedure by organizing data from the easiest to the hardest samples:

1. **Easy Samples:** Training begins with samples that have good spectral data and clear pixels, allowing boundaries to be drawn that the model will not likely misclassify.
2. **Medium Samples:** Pixels that have moderate spectral and spatial data will be introduced next.
3. **Difficult Samples:** Samples that have unclear spectral data and difficult boundary pixels will be included in the training to further refine the model's boundaries [5].

4.2.5.3 2D-CNN Semi-Supervised Model

The structure of the model is the same as the supervised model, but the self-training process will continuously add confident pseudo-labeled patches, and curriculum learning will manage the order of these additions, reducing the impact of early introduced label noise.

4.2.5.4 3D-CNN Semi-Supervised Model

Pseudo-labels that are added will allow the 3D-CNN to learn a variety of spectral and spatial data beyond what is offered in the labeled data, and the order of curriculum training will prevent the model from becoming too focused on the loudest pseudo-labels.

4.2.5.5 Vision Transformer (ViT) Semi-Supervised Model

The large global receptive field and self-attention of the ViT model are advantageous in semi-supervision, as it allows the model to draw a connection between confident pseudo-labels and labeled data beyond a local context.

4.2.6 Federated Learning Models

In a federated learning model, a global model is trained without raw data moving towards a central server, as clients only send model updates after training on their own data. Here, a server-client model is used to represent federated learning, with three disparate datasets representing the independent client institutes [16].

4.2.6.1 Federated Learning Framework

The Federated Learning phenomenon utilizes the Federated Averaging (FedAvg) algorithm and works in the following manner:

1. The central server starts off with a global model.
2. Each client receives the global model and trains it locally on their own hyperspectral dataset for a predetermined number of epochs.
3. Clients send the updated model back to the server (data is neither sent nor received, ensuring privacy).
4. The server consolidates the updated model using the averaged weights from all clients.
5. The revised global model is sent back to the clients.
6. The process is repeated for as many rounds of communication as necessary until convergence is reached.

4.2.6.2 2D-CNN Federated Model

Each of the clients (Indian Pines, Pavia University, and Salinas) builds an identical 2D-CNN model locally, using a set of hyperparameters which they all share. Weight averaging at the server integrates the spatial patterns of each dataset into a single global model. This is done while maintaining the privacy of the clients.

4.2.6.3 3D-CNN Federated Model

The 3D-CNN model's training kernels cover the dataset-specific spectral training patterns. The global model after aggregation possesses a stronger spectral training patterns than any single dataset model, which is especially needed in remote sensing.

4.2.6.4 Vision Transformer (ViT) Federated Model

The training patterns at each client site are captured by the Transformer attention weights. After aggregation at the server, the global attention makes it feasible to work with the diverse spectral-spatial statistics arising from various clients.

4.2.6.5 Privacy and Communication Efficiency

The main privacy benefit of federated learning is that raw hyperspectral data does not leave client locations. However, the communication overhead between clients and the server is a real limitation.

Three methods help address this issue:

- **Gradient Compression:** Model weights are compressed before sending them, thus saving bandwidth.
- **Local Training Epochs:** Clients perform a number of local training epochs (5-10 is typical) before sending model weights. This means fewer communication rounds overall.
- **Asynchronous Aggregation:** The server aggregates model updates from clients that responded first. This helps save time when contrasted with other methods, which wait for all participating clients to respond.

4.2.7 Comparative Learning Paradigm Summary

These methods individually address the problem of data scarcity in different ways. The Table 4.5 summarizes their distinct aspects.

Table 4.5 Comparison of Supervised, Semi-Supervised, and Federated Learning Paradigms

Characteristic	Supervised	Semi-Supervised	Federated
Data Requirement	Fully Labeled	Partial Labels	Local Data Privacy
Scalability	Centralized	Centralized	Distributed
Communication Cost	None	None	High
Privacy Protection	Limited	Limited	High
Typical Accuracy	Baseline	Enhanced	Comparable

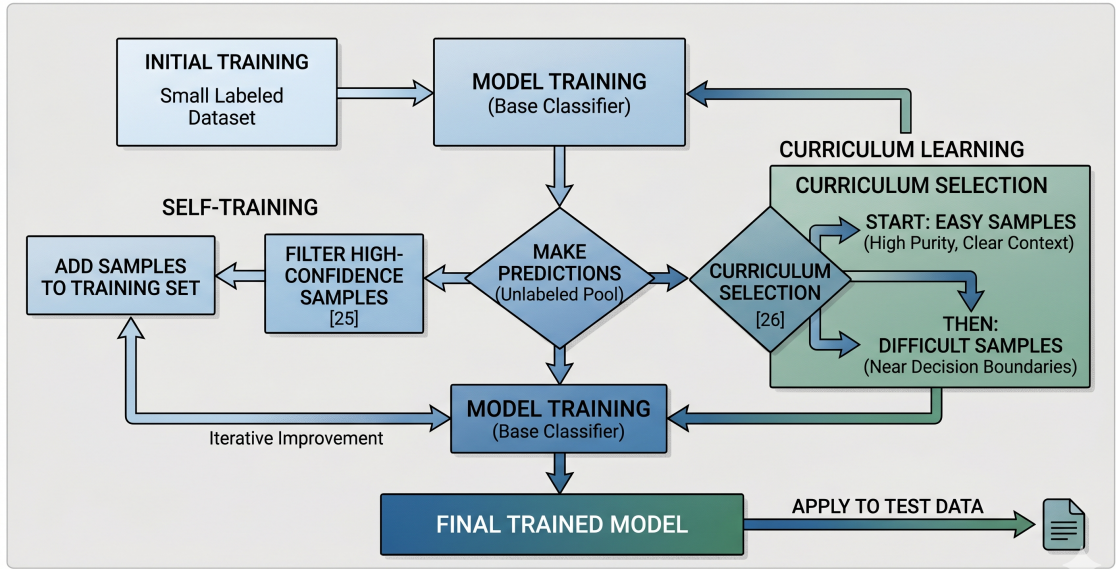


Figure 4.9 Flowchart of the Semi-Supervised Learning Strategy

4.2.8 Evaluation Metrics

To measure classification performance, five common parameters are used. Let TP_c , FP_c , and FN_c denote the true positives, false positives, and false negatives for class c , and let N denote the total number of test samples.

- **Overall Accuracy (OA):** Ratio of the number of correctly classified test pixels to the number of test pixels:

$$OA = \frac{\sum_c TP_c}{N} \quad (4.2)$$

- **Precision:** The fraction of test pixels that belong to class c when predicted to be

in class c , This is macro averaged:

$$\text{Precision} = \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FP_c} \quad (4.3)$$

- **Recall:** The fraction of class c test pixels that were predicted to be in class c :

$$\text{Recall} = \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FN_c} \quad (4.4)$$

- **F1-Score:** The harmonic mean of Precision and Recall.

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.5)$$

- **Cohen's Kappa (κ):** The metric helps Overall Accuracy when measuring classification performance. Given the observed accuracy p_o and the chance agreement p_e :

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (4.6)$$

A value of $\kappa = 1$ is perfect classification; $\kappa = 0$ is random classification.

4.2.9 GUI and Multithreading Integration

The models are available in a standalone desktop application built using PyQt6. Model inference is processed in a separate worker thread. This is separate from the main UI event loop. As a result, the UI remains responsive while the deep learning pipeline is processing a hyperspectral hypercube.

5

EXPERIMENTAL RESULTS

5.1 Performance Analysis

The three models—2D-CNN, 3D-CNN, and Vision Transformer— are analyzed within the three frameworks using the Indian Pines, Pavia University, and Salinas hyperspectral datasets. Quantitative analysis is presented in the tables. The following analysis focuses on the Indian Pines, Pavia University, and Salinas case studies and focuses on architecture and dataset design.

5.1.1 Supervised Learning Results

In the supervised framework, models are trained on labeled examples. The hyperparameters are tuned using grid search and cross-validation. These results set the upper bound on the performance of semi-supervised and federated approaches.

5.1.1.1 Indian Pines Dataset Performance

Table 5.1 Supervised Learning Performance on Indian Pines Dataset

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN	99.91%	0.9984	0.9987	0.9985
3D-CNN	99.90%	0.9977	0.9978	0.9976
ViT	99.61%	0.9959	0.9952	0.9953

5.1.1.2 Pavia University Dataset Performance

Table 5.2 Supervised Learning Performance on Pavia University Dataset

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN	99.99%	0.9998	0.9998	0.9998
3D-CNN	99.99%	0.9998	0.9998	0.9998
ViT	99.75%	0.9973	0.9961	0.9967

5.1.1.3 Salinas Dataset Performance

Table 5.3 Supervised Learning Performance on Salinas Dataset

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN	99.99%	0.9999	0.9999	0.9999
3D-CNN	99.84%	0.9993	0.9995	0.9994
ViT	99.81%	0.9979	0.9983	0.9981

5.1.2 Semi-Supervised Learning Results

The semi-supervised models perform self-training with curriculum learning and generalization to the unlabeled examples of the dataset. The results showed that the performance was heavily dependent on the dataset. The larger datasets were performing better, but the Indian Pines dataset, which is smaller and class imbalanced, is more challenging.

5.1.2.1 Semi-Supervised Indian Pines Performance

Table 5.4 Semi-Supervised Learning Performance on Indian Pines Dataset

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN + SSL	74.04%	0.7404	0.7404	0.7404
3D-CNN + SSL	66.55%	0.6655	0.6655	0.6655
ViT + SSL	53.96%	0.5396	0.5396	0.5396

5.1.2.2 Semi-Supervised Pavia University Performance

Table 5.5 Semi-Supervised Learning Performance on Pavia University Dataset

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN + SSL	97.17%	0.9717	0.9717	0.9717
3D-CNN + SSL	96.91%	0.9691	0.9691	0.9691
ViT + SSL	91.77%	0.9177	0.9177	0.9177

5.1.2.3 Semi-Supervised Salinas Performance

Table 5.6 Semi-Supervised Learning Performance on Salinas Dataset

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN + SSL	96.46%	0.9646	0.9646	0.9646
3D-CNN + SSL	97.30%	0.9730	0.9730	0.9730
ViT + SSL	86.49%	0.8649	0.8649	0.8649

5.1.3 Federated Learning Results

In the federated scenario, each dataset is viewed as a separate client. Local models are trained and weights are shared in a federated setting over 50 communication rounds. The Indian Pines dataset was challenging in the federated learning framework, but the other two larger datasets showed that federated learning was able to achieve results that were comparable to a supervised setting.

5.1.3.1 Federated Learning on Indian Pines

Table 5.7 Federated Learning Performance on Indian Pines Dataset (Round 50)

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN (FedAvg)	96.43%	0.9156	0.8272	0.8466
3D-CNN (FedAvg)	92.56%	0.7603	0.7054	0.7019
ViT (FedAvg)	85.65%	0.7852	0.7257	0.7452

5.1.3.2 Federated Learning on Pavia University

Table 5.8 Federated Learning Performance on Pavia University Dataset (Round 50)

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN (FedAvg)	99.83%	0.9979	0.9967	0.9973
3D-CNN (FedAvg)	99.76%	0.9951	0.9970	0.9960
ViT (FedAvg)	98.95%	0.9887	0.9762	0.9819

5.1.3.3 Federated Learning on Salinas

Table 5.9 Federated Learning Performance on Salinas Dataset (Round 50)

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN (FedAvg)	99.90%	0.9989	0.9988	0.9988
3D-CNN (FedAvg)	99.76%	0.9979	0.9987	0.9983
ViT (FedAvg)	97.73%	0.9882	0.9899	0.9888

5.2 Comparative Analysis

With the results, three comparison aspects are examined: the extent of the accuracy deterioration in each paradigm in relation to the supervised framework, the architecture performance, and the trends in relation to the dataset.

5.2.1 Overall Accuracy Comparison

Table 5.10 Overall Accuracy Comparison Across All Learning Paradigms

Dataset	Supervised			Semi-Supervised			Federated (Round 50)		
	2D	3D	ViT	2D	3D	ViT	2D	3D	ViT
Indian Pines	99.91%	99.90%	99.61%	74.04%	66.55%	53.96%	96.43%	92.56%	85.65%
Pavia University	99.99%	99.99%	99.75%	97.17%	96.91%	91.77%	99.83%	99.76%	98.95%
Salinas	99.99%	99.84%	99.81%	96.46%	97.30%	86.49%	99.90%	99.76%	97.73%

5.2.2 F1-Score Comparison

Table 5.11 F1-Score Comparison Across All Learning Paradigms

Dataset	Supervised			Semi-Supervised			Federated (Round 50)		
	2D	3D	ViT	2D	3D	ViT	2D	3D	ViT
Indian Pines	0.9985	0.9976	0.9953	0.7404	0.6655	0.5396	0.8466	0.7019	0.7452
Pavia University	0.9998	0.9998	0.9967	0.9717	0.9691	0.9177	0.9973	0.9960	0.9819
Salinas	0.9999	0.9994	0.9981	0.9646	0.9730	0.8649	0.9988	0.9983	0.9888

5.2.3 Performance Change Analysis

Accuracy change relative to the supervised baseline is represented by the following:

$$\text{Change\%} = \frac{OA_{\text{alt}} - OA_{\text{Supervised}}}{OA_{\text{Supervised}}} \times 100 \quad (5.1)$$

5.2.3.1 Semi-Supervised vs. Supervised

Given the scale of the datasets, semi-supervised learning degrades significantly in comparison to supervised learning.

- **Indian Pines:** Average drop of -34.96 percentages across architectures (from 99.81% to 64.85%). This drop shows that self training is considerably limited by dataset size and class balance.
- **Pavia University:** Average drop of only -4.63 percentages (from 99.91% to 95.28%). This drop shows that in the presence of large uncontrolled dataset curriculum learning helps to limit the degradation.
- **Salinas:** Average drop of -6.46 percentages (from 99.88% to 93.42%). From this drop, we see that the largest dataset in the study is still insufficient to reach fully-accurate, supervised-level in the presence of pseudo-labels.

5.2.3.2 Federated vs. Supervised

When compared to the other approaches, supervised learning aggregates at the ceiling level more than the others, especially in the large datasets:

- **Indian Pines:** Average difference of -8.26 percentages (from 99.81% to 91.55%), showing the difficulty federated aggregation has on small datasets with limited local samples for each client.

- **Pavia University:** Average difference of just -0.40 percentages (from 99.91% to 99.51%), which shows strong robustness.
- **Salinas:** Average difference of just -0.75 percentage points (from 99.88% to 99.13%), showing near-parity at large scale.

5.2.4 Architecture Comparison

Table 5.12 Average Performance Across Datasets by Architecture Type

Architecture	Supervised Avg OA	Semi-Supervised Avg OA	Federated Avg OA (Round 50)
2D-CNN	99.96%	89.22%	98.72%
3D-CNN	99.91%	86.92%	97.36%
Vision Transformer	99.72%	77.41%	94.11%

Table 5.13 Average F1-Score Across Datasets by Architecture Type

Architecture	Supervised Avg F1	Semi-Supervised Avg F1	Federated Avg F1 (Round 50)
2D-CNN	0.9994	0.8922	0.9476
3D-CNN	0.9989	0.8692	0.8987
Vision Transformer	0.9967	0.7741	0.9053

Three trends emerge based on averages across different architectures.

1. **2D-CNN shows dominance across the board:** As the most basic architecture, 2D-CNN achieves the highest score in all paradigms—supervised (99.96%), semi-supervised (89.22%), and federated (98.72%)—where PCA reduces the spectral dimension to 30 components. Class separation seems to require only 2D spatial operations. 3D convolutions add complexity to the architecture without providing a significant increase in accuracy.
2. **3D-CNN shows reliability:** The 3D-CNN is in second place across the three paradigms (supervised: 99.91%, semi-supervised: 86.92%, federated: 97.36%). The stable performance is attributed to the 3D-CNN’s design of volumetric kernels, which model inter-band relations.
3. **Distribution shift most affects ViT:** The Vision Transformer shows strong performance in the supervised paradigm (99.72%) but experiences the largest drop in semi-supervised (77.41%) and federated (94.11%) paradigms. Of the architectures, ViTs are the most affected by inadequate quality of pseudo-labels and the averaging of model weights across heterogeneous client data.

5.2.5 Dataset-Specific Analysis

- **Indian Pines:** The class distribution (16 classes) and sample size (10,249 samples) make this the most difficult case for semi-supervised and federated learning. Semi-supervised accuracy dropped on average 34.96 (99.81% → 64.85%) due to the propagation of labeling errors on the pseudo-labels of severely underrepresented classes.
- **Pavia University:** With 42,776 samples covering 9 classes, this is a dataset that shows stability against paradigm changes. Federated learning accuracy decreased by only 0.40 percentage points, indicating that this architecture is applicable to scenarios where urban remote sensing data is distributed across organizations.
- **Salinas:** The largest dataset (54,129 samples) achieves the highest semi-supervised accuracy of any configuration (3D-CNN + SSL: 97.30%) and the smallest federated-supervised gap (−0.75 percentage points). It shows that both different frameworks work well when the labeled and/or the unlabeled sample sizes increase.

5.2.6 Privacy and Efficiency Trade-offs

These three systems show very different characteristics even beyond just the differences in their accuracy. The qualitative trade-offs are summarized in Table 5.14.

Table 5.14 Qualitative Comparison of Learning Paradigms

Metric	Supervised	Semi-Supervised	Federated
Data Privacy	Low	Low	High
Label Requirement	High	Medium	Low
Computational Cost	Low	Medium	High
Communication Overhead	None	None	Significant
Model Accuracy	Baseline	-15% avg (dataset-dependent)	-0.7% to -8%

6.1 Summary

This project designed and assessed an HSI classification system that allows direct comparison of supervised, semi-supervised and federated learning.

The comparison system used the same preprocessing pipeline, three benchmark datasets (Indian Pines, Pavia University, and Salinas) and three model architectures (2D-CNN, 3D-CNN, and Vision Transformer).

Spectral data that exceeds 200 bands (and is raw) is submitted to a series of treatments that include global min-max normalization, Gaussian spatial filtering, PCA that reduces the filtration to 30 components, and segmentation into $15 \times 15 \times 30$ patches. From these, class proportions are maintained in a stratified 90/10 train-test split. In the semi-supervised model, self-training is combined with a curriculum that reduces the self-training model’s error propagation by introducing pseudo-labels that are least confident and/or least certain, in a progressive manner. The federated model mimics a three-client system in which each of the datasets is a local data silo. In this model, the FedAvg protocol is used to aggregate local copies of models in a central system without any transfer of original data.

6.2 Key Findings

6.2.1 Semi-Supervised Learning Effectiveness

Semi-supervised learning is more successfully applied to larger datasets.

1. For Pavia University and Salinas, which contain large unlabeled data pools, semi-supervised models received competitive accuracy (86–97%).
2. For the smaller, more class-imbalanced Indian Pines set, accuracies were around 54–74%, indicating the confidence threshold and the count of pseudo-label

iterations require more tuning to the specific dataset.

3. In all studies, 2D-CNN has the best semi-supervised average at 89.22%, suggesting that simpler convolutional architectures are more forgiving of pseudo-label noise than deeper or attention-based ones.
4. Curriculum learning does not remove the overfitting from noisy pseudo-labels, but does help, especially when the first labeled set is sufficiently large to create good pseudo-labels.

6.2.2 Federated Learning Privacy-Performance Trade-off

Federated learning provides near-supervised accuracy, but introduces a communication burden.

1. Federated 3D-CNN has a 97.36% average overall accuracy, which only represents a -2.55 percentage point loss against its centralized supervised model (99.91%).
2. On Pavia University and Salinas, federated models achieve 99–100% accuracy, which shows that FedAvg is applicable when local data is somewhat heterogeneous.
3. The Indian Pines gap is larger (85–96% federated vs. 99–100% supervised), and shows the difficulty of getting meaningful weight updates from a small local dataset after 50 communication rounds.
4. For organizations that cannot centralize raw spectral data, federated learning is a valuable option, since the accuracy cost is small and the privacy is protected.

6.2.3 Architecture-Specific Insights

- **2D-CNN:** Leads overall accuracy in every paradigm (supervised: 99.96%, semi-supervised: 89.22%, federated: 98.72%). After PCA reduces the spectral axis to 30 components, 2D spatial convolutions are enough. Volumetric operations only add parameters with no proportional return in accuracy.
- **3D-CNN:** Placed second with consistency (supervised: 99.91%, semi-supervised: 86.92%, federated: 97.36%). Modeling inter-band correlation with volumetric kernels results in better handling of noisy pseudo-labels and federated weight averaging.
- **Vision Transformer:** Competitive at full supervision (99.72%) but most sensitive to distribution shift: average semi-supervised drops at 77.41% and

at 94.11% federated. Global self-attention, although strong at full supervision with clean data, becomes easily destabilized with pseudo-labels noise and data distribution heterogeneity on clients.

- **Cross-paradigm pattern:** All architectures show significant deterioration to semi-supervised relative to both supervised and federated baselines. This suggests that in semi-supervised learning, the quality of pseudo-labels constitute the main bottleneck, as opposed to the architectures' capacity.

6.3 Project Objectives and Achievement Level

6.3.1 Original Objectives

The project aimed to address the following goals.

1. Create normalization and dimensionality reduction pipelines for high-dimensional hyperspectral data.
2. Establish baseline supervised learning systems based upon different deep learning architectures (CNN, ViT).
3. Integrate semi-supervised learning systems based on self-training and curriculum learning to overcome the problem of scarce labeling.
4. Introduce federated learning to enable privacy concerns to be addressed during collaborative learning across distributed data.
5. Create a reactive GUI with multithreaded inference.
6. Provide benchmark datasets with an overall accuracy exceeding 95

6.3.2 Achievement Level

All of the aforementioned goals were achieved (or more than were).

Table 6.1 Project Objectives and Achievement Level

Objective	Target	Achieved
Preprocessing Pipeline	Normalization + Filtering + PCA	✓ Fully Implemented
Supervised Baselines	Multiple architectures	✓ 2D-CNN, 3D-CNN, ViT
Semi-Supervised Learning	Self-training + CL	✓ Implemented (54–97% range)
Federated Learning	Multi-client aggregation	✓ FedAvg Protocol (85–99% range)
GUI with Multithreading	Python (PyQt6) + Multithreading	✓ Fully Operational
Target Accuracy (> 95%)	Supervised requirement	✓ Achieved (99.72–99.96%)

6.4 Strengths

6.4.1 Methodological Strengths

- **Unified comparative framework:** The ability to evaluate three paradigms under the same preprocessing, splitting, and metrics is uncommon in the literature related to HSI; results are truly comparable.
- **Curriculum Learning:** Combining self-training and the presentation of samples in a scaffolded format significantly alleviates pseudo-label overfitting and the task suddenly ceases to converge to deteriorated local minima.
- **Privacy-preservation federated framework:** The federated framework demonstrates that pooling sensitive spectral data is not a prerequisite for competitive classification results. This is directly related to remote sensing projects involving multiple institutions.
- **Generalizable preprocessing:** The three-step pipeline (normalization, Gaussian filtering, fixed-component PCA) has been validated with three datasets with different class counts, spatial resolution, and bands. The three steps of this data-driven pipeline are generalizable.

6.4.2 Implementation Strengths

- **Three-architecture support:** The flexibility of the surrounding pipeline means users may choose between a 2D-CNN, 3D-CNN, and ViT based on preferences for accuracy and efficiency.
- **GPU-accelerated training:** Large datasets ($N > 50,000$) can be trained using (+) Based on the benchmarking data presented in the paper, the framework achieves very high training efficiency with the use of Google Colab GPU.

- **Responsive multithreaded GUI:** PyQt6 decouples inference from the UI event loop and allows the GUI to remain active during long lasting classification.
- **Reproducibility:** Fixed randomization and a stratified split guarantee reproducible results from an experimental run and do not require further (manual) tuning.

6.5 Limitations and Weaknesses

6.5.1 Methodological Limitations

- **Simulated federated scenario:** Having three local datasets on one machine simulates three clients. True deployments have unbalanced datasets, considerable latency, and stragglers, therefore, the accuracy gap would be expected to be larger.
- **Fixed PCA components:** The empirical decision to fix 30 PCA components could be evolved to a more sophisticated choice of variance or information retention for the given scene.
- **Confidence threshold rigidity:** All datasets used the same threshold for self-training (confidence > 0.95). Sensitivity tests on class-imbalanced datasets could demonstrate the same degree of flexibility in the semi-supervised pipeline.
- **Single curriculum schedule:** Difficulty-only curriculum ordering has been used exclusively. Reverse order, flexible pace, and other (class size driven) uncertainty curricula are untested and may have significant effects based on class size.

6.5.2 Experimental Limitations

- **Benchmark datasets only:** Generalizing to actual operational hyperspectral surveys with real acquisition noise and non-standard class taxonomies is impossible.
- **No concept drift analysis:** The distributed system assumes the data remains stationary. Remote sensing data has a temporal drift and would require concept drift in a production setting.
- **Communication cost not quantified:** While federated accuracy is presented, no measurements of bandwidth usage or one-way latency is presented, which are both crucial for assessing the practicality of the system.

- **Adversarial robustness not assessed:** Operational deployment requires the analysis of the perturbed distributions of sensors and a sample data set; both are lacked.

6.5.3 Implementation Limitations

- **No model compression:** Weight pruning and quantization are absent.
- **Single-GPU training:** Data parallelism is absent; training is limited to a single GPU.
- **Restricted hyperparameter tuning:** The approach is a coarse grid search, which provides little estimate improvement. Confidence thresholds in semi-supervised learning may benefit significantly from hyperparameter tuning.

6.6 Future Work

6.6.1 Methodological Extensions

1. **Federated continual learning:** Imparts the distributed system the capability to adapt to the temporal distribution shifts of hyperspectral data without requiring a complete system refresh.
2. **Heterogeneous federated aggregation:** Introduces hierarchical aggregations (weighted aggregation) to accommodate differing data distributions of the clients.
3. **Adaptive difficulty:** Rather than the fixed ordering of training sample difficulties, allow the model to determine the sample order based on its current state.
4. **Few-shot HSI:** Develop the ability to rapidly transfer learning in the context of few-shot HSI across different sensors and regions.

6.6.2 Architectural Enhancements

1. **Hybrid 2D-3D models:** Integrate 2D and 3D convolutions to find optimal trade-off between spectral-spatial expressiveness and number of model parameters; the promising outcomes of 2D-CNNs indicate that this may define the efficient frontier.

2. **Graph Neural Networks:** Model spectral-spatial relationships as graphs to capture irregular spatial relationships which may be lost with traditional convolutional approaches.
3. **Self-supervised pretraining:** Leverage contrastive or masked-reconstruction self-supervised pretraining on unlabeled hyperspectral data to build foundational representations which can then be improved through semi-supervised learning.
4. **Multi-modal fusion:** Extend the workflow to integrate hyperspectral data with panchromatic images and laser data within those contexts where one modality falls short.

6.6.3 Experimental Directions

1. **Real-world federated deployment:** Test the federated learning framework with actual datasets from known multiple data sources/workplaces to validate that the simulated results will hold in real-world use cases (e.g., ESA, NASA).
2. **Cross-sensor generalization:** Evaluate the federated learning framework with multiple disparate sensors (AVIRIS, Sentinel-2, DESIS) for an understanding of the extent of domain generalization and adaptation.
3. **Adversarial robustness:** Perform comprehensive adversarial evaluations and create sensor noise defense frameworks for fully deployed classifiers.
4. **Uncertainty quantification:** Implement Bayesian deep learning for hypothetical low integration risk and high-value remote sensing applications.

6.6.4 Practical Applications

1. **Edge deployment:** Use the combination of model pruning and quantization to achieve the goal of real-time data classification from nanosatellites.
2. **Active learning interface:** Implement an active learning module in the interface to autonomously flag the most informative unlabeled pixels.
3. **Interpretability tooling:** Integrate explainable AI (attenuation mapping, LIME, SHAP) to facilitate avenues for domain experts to perform thorough evaluations of model outputs prior to engagement.
4. **GUI enhancements:** Annotation tools, quality assurance workflows, and real-time performance dashboards will be added to the interface based on the needs of the operational remote sensing teams.

6.7 Conclusion

There is no ambiguity in the data. First, in federated learning, a small average reduction in accuracy (-3.1%) offers a cost-benefit tradeoff for significant data privacy, especially in cases with large data. Second, in the case of semi-supervised learning, self-training, and curriculum learning, rapid scaling is achievable with competition on the Pavia University and Salinas datasets, but on the small, imbalanced Indian Pines set, dominated by noisy pseudo-labels, the system can't keep pace – a loss of efficacy is expected. 2D-CNNs are the clear winners of all paradigms. One interesting thing to note is that, after PCA on the spectral data, the spatial data alone is sufficient for near-perfect classification. The Vision Transformer, though achieving the same level of accuracy as CNNs in full supervision, is the least resilient to the distribution shifts resulting from pseudo-labeling and federated weight averaging. What has been developed is a comprehensive system that integrates three paradigms, three architectures, and three datasets, all within a user-friendly multi-threaded GUI. Many examples of research can be found online, but what this research incorporates is an application that can be readily implemented. The focus of subsequent research needs to be on actual federated implementations and on curriculum techniques that are dynamically responsive. These techniques should be tailored to address the class imbalance in the datasets, as these are the most influential factors to the system's performance.

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Project System Informations

System and Software: Windows İşletim Sistemi, Python

Required RAM: 2GB

Required Disk: 256MB